STAT 516: Directed Acyclic Graphs and Structural Equation Models

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Reading guide

English transcription of `notes/stat516-note2.pdf` (October 15, 2026), checked against Yen-Chi Chen's supplied *Lecture 2: Graphical Models* (L2), Section 2.4. Source-page labels refer to the handwritten PDF. Yellow boxes retain TODOs, red boxes explain corrections, and blue boxes supply important context. Source page 8 is blank.

1 Directed acyclic graphs

Handwritten source: 1–2; L2, Section 2.4.

A directed acyclic graph (DAG) is a directed graph $G = (V, E)$ with no directed cycle. Using the convention $E_{ij} = 1$ for $X_i \to X_j$, the parent and child sets are

$$\mathrm{PA}(i) = \{j : E_{ji} = 1\}, \tag{1}$$

$$\mathrm{Ch}(i) = \{j : E_{ij} = 1\}. \tag{2}$$

We use vertex indices in these sets; $X_{\mathrm{PA}(i)}$ denotes the corresponding random vector. The reverse edge is absent if $E_{ij} = 1$, but that restriction alone does not rule out longer directed cycles. A density factorizes according to the DAG when

$$p(x_1, \ldots, x_d) = \prod_{i=1}^{d} p(x_i \mid x_{\mathrm{PA}(i)}). \tag{3}$$

For a node with no parents, its factor is its marginal density. Together with these conditional distributions, the DAG defines a Bayesian network and a generative process.

#TODO: Complete the DAG-factorization explanation (source p. 2)

The handwritten Chinese word means "supplement" or "fill in" and does not specify a narrower question. A natural completion is why acyclicity matters. A finite DAG has a topological ordering, in which all parents precede their children. Sample each variable from its conditional distribution in that order. The product above integrates or sums to one: eliminate variables in reverse topological order, so each last remaining conditional factor integrates to one. Cyclic conditional specifications do not generally define a normalized joint distribution by this product. The DAG alone specifies a family of models; the conditional distributions must also be supplied.

1.1 A five-variable generative model

Handwritten source: 1 and 3; compare L2, Figure 2.1.

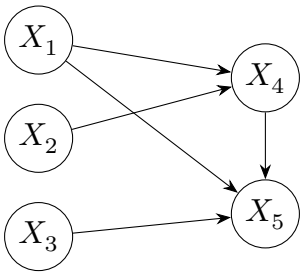
Take the root variables to be mutually independent, with

$$X_1 \sim N(0,1), \qquad X_2 \sim \mathrm{Unif}[0,1], \qquad X_3 \sim N(0,4), \tag{4}$$

$$X_4 \mid X_1, X_2 \sim N(X_1 + X_2^2, 1), \tag{5}$$

$$X_5 \mid X_1, X_3, X_4 \sim \mathrm{Unif}(0, X_1^2 + 2X_3^2 + e^{X_4}). \tag{6}$$

The upper endpoint of the last interval is strictly positive. The conditional specifications are interpreted as depending only on the listed parents. The graph is



Its joint density is

$$p(x) = p_1(x_1)p_2(x_2)p_3(x_3)p(x_4 \mid x_1, x_2)p(x_5 \mid x_1, x_3, x_4). \tag{7}$$

Correction: The parents of X_5 (source pp. 1 and 3)

The handwritten generative specification and arrows make the parents of X_5 equal to X_1, X_3, X_4. Several handwritten factors instead use X_1, X_2, X_4. Those occurrences of X_2 are replaced with X_3. The independent roots are an explicit model assumption consistent with the DAG factorization; their marginal laws alone would not imply independence.

The resulting conditional-independence statement is

$$p(x_5 \mid x_1, x_2, x_3, x_4) = p(x_5 \mid x_1, x_3, x_4), \tag{8}$$

$$X_5 \perp\!\!\!\perp X_2 \mid (X_1, X_3, X_4). \tag{9}$$

Supplement: The DAG local Markov property

A node is conditionally independent of its non-descendants, excluding its parents, given its parents. This is the directed local Markov property; it is distinct from conditioning on all neighbors in an undirected graph.

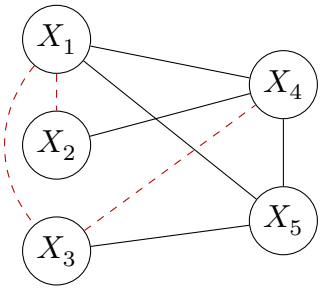
2 Moralization and conditional independence

Handwritten source: 3–5; L2, Definition 2.22 and Figures 2.1–2.3.

To construct the moral graph of a DAG, join every pair of parents of each child and replace all arrows by undirected edges. For the five-variable example, the original skeleton has edges

$$\{1,4\},\ \{2,4\},\ \{1,5\},\ \{3,5\},\ \{4,5\}. \tag{10}$$

The additional moral edges are $\{1,2\}$, $\{1,3\}$, and $\{3,4\}$.



Dashed red edges join co-parents.

Each DAG factor involves a child and its parents, which form a clique in the moral graph. Hence the distribution factorizes over the moral graph. Every separation in the moral graph therefore implies a conditional independence for a distribution that factorizes over the original DAG. Some DAG independences are lost in this representation: for example, $X_1 \perp\!\!\!\perp X_2$ follows from the independent roots, but their moral edge hides it.

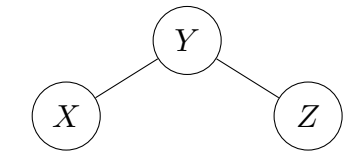
2.1 Four three-variable configurations

Handwritten source: 4–5; compare L2, Sections 2.4.1 and 2.4.3.

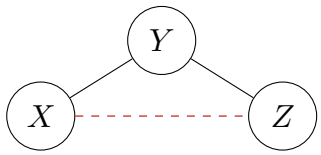
The handwritten diagrams can be written as follows:

$$G_1: \quad X \leftarrow Y \rightarrow Z, \qquad p(x,y,z) = p(y)p(x \mid y)p(z \mid y), \tag{11}$$
$$G_2: \quad X \rightarrow Y \rightarrow Z, \qquad p(x,y,z) = p(x)p(y \mid x)p(z \mid y), \tag{12}$$
$$G_3: \quad X \rightarrow Y \leftarrow Z, \qquad p(x,y,z) = p(x)p(z)p(y \mid x,z), \tag{13}$$
$$G_4: \quad X \leftarrow Y \leftarrow Z, \qquad p(x,y,z) = p(z)p(y \mid z)p(x \mid y). \tag{14}$$

The first, second, and fourth graphs imply $X \perp\!\!\!\perp Z \mid Y$. The collider graph G_3 implies $X \perp\!\!\!\perp Z$; conditioning on Y can induce dependence.



Moral graph of G_1, G_2, G_4



Moral graph of G_3

Correction: Skeletons, moral graphs, and dependence (source pp. 4–5)

All four DAGs have the same undirected *skeleton*, the path $X - Y - Z$. They do not all have the same moral graph. For the collider G_3, moralization joins its parents X and Z, giving a triangle. The handwritten triangular moral graph cannot describe all four cases. An open path permits dependence; it does not force dependence for every parameter choice. Thus the claim $X \not\perp\!\!\!\perp Z \mid Y$ for a collider is generally true for typical parameters, but is not a universal consequence of factorization alone. An unfaithful model can have additional independences.

Supplement: A concrete collider calculation

Let X, Z be independent $\mathrm{Ber}(1/2)$ variables and set $Y = X + Z$. Then $X \perp\!\!\!\perp Z$, but conditional on $Y = 1$,

$$\mathbb{P}(X = 1, Z = 1 \mid Y = 1) = 0 \neq \mathbb{P}(X = 1 \mid Y = 1)\mathbb{P}(Z = 1 \mid Y = 1) = \frac{1}{4}. \tag{15}$$

This demonstrates the dependence that observing a common effect can create.

2.2 What observational data can identify

The DAGs G_1, G_2, G_4 have the same d-separation statements and are Markov equivalent. Conditional-independence information alone cannot distinguish them. The collider has a different independence pattern and can be distinguished under suitable Markov and faithfulness assumptions.

Correction: Data do not determine only an undirected graph

The handwritten statement "data could only give UG" is too broad. Under suitable assumptions, observational conditional independences can identify a Markov equivalence class, including some arrowheads. They generally cannot identify a unique DAG. Additional domain knowledge, interventions, or identifiable structural assumptions may resolve directions. This qualifies the identifiability warning in L2, pp. 2-13–2-14.

Supplement: D-separation in brief (L2, Section 2.4.1)

A path is blocked by conditioning on a non-collider along that path. A collider blocks a path unless the collider or one of its descendants is conditioned on. Two vertex sets are d-separated given a conditioning set if every path between them is blocked. D-separation implies conditional independence for a distribution that factorizes according to the DAG. For an exact test using moralization, first restrict to the ancestors of the queried variables and the conditioning variables, then moralize and check separation; moralizing the entire DAG may hide valid independences.

2.3 Causal interpretation

Handwritten source: 5; L2, Section 2.4.3.

In a *causal* DAG, $X \to Y$ represents a direct causal relationship relative to the modeled variables. An ordinary Bayesian-network factorization does not by itself establish this causal meaning. A causal interpretation requires assumptions about the data-generating mechanisms, and conditioning on a variable is generally different from intervening on it.

3 Hierarchical Bayesian models

Handwritten source: 6; L2, Section 2.4.2.

Let X_i be student i's exam score and θ_i the student's latent ability. A simple model is

$$X_i \mid \theta_i \sim N(\theta_i, \sigma^2), \tag{16}$$

$$\theta_i \sim N(\mu, \tau^2), \tag{17}$$

with independent abilities and conditionally independent scores given the abilities. The parameters μ, τ^2 describe the population. For a group or occasion indexed by t, the handwritten extension introduces a random group mean:

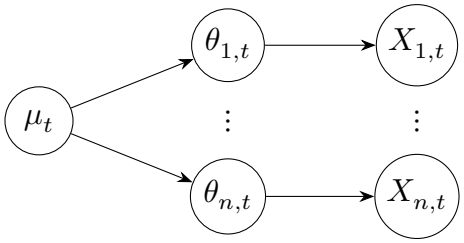
$$\mu_t \sim N(\omega, \delta^2), \tag{18}$$

$$\theta_{i,t} \mid \mu_t \sim N(\mu_t, \tau^2), \qquad i = 1, \dots, n, \tag{19}$$

$$X_{i,t} \mid \theta_{i,t} \sim N(\theta_{i,t}, \sigma^2). \tag{20}$$

Abilities are conditionally independent given μ_t, and scores are conditionally independent given their respective abilities. This gives

$$p(\mu_t, \theta_{1:n,t}, x_{1:n,t}) = p(\mu_t) \prod_{i=1}^{n} p(\theta_{i,t} \mid \mu_t) p(x_{i,t} \mid \theta_{i,t}). \tag{21}$$



Supplement: Shared latent variables induce dependence

For distinct students in the same group, integrating out the random group mean gives

$$\mathrm{Var}(X_{i,t}) = \delta^2 + \tau^2 + \sigma^2, \tag{22}$$

$$\mathrm{Cov}(X_{i,t}, X_{j,t}) = \delta^2 \qquad (i \neq j). \tag{23}$$

Conditional independence given μ_t need not imply marginal independence. Here t labels the groups or occasions; the handwriting does not specify temporal dependence between different groups.

4 Structural equation models

Handwritten source: 6–7; L2, Section 2.4.4.

A structural equation model (SEM) specifies

$$X_i = f_i(X_{\mathrm{PA}(i)}, \varepsilon_i), \qquad i = 1, \dots, d, \tag{24}$$

where the exogenous noise variables are mutually independent. Acyclicity allows the equations to be evaluated in topological order. Their induced conditional distributions yield the DAG factorization. An additive-noise model is the special case

$$X_i = m_i(X_{\mathrm{PA}(i)}) + \varepsilon_i. \tag{25}$$

A linear SEM further assumes

$$X_i = \gamma_{i,\mathrm{PA}(i)}^T X_{\mathrm{PA}(i)} + \varepsilon_i, \tag{26}$$

with an optional intercept. Normal noise is an additional assumption, not part of the general SEM definition.

Correction: General SEM versus Gaussian additive noise (source p. 6)

The handwritten transition from f_i to an additive Gaussian-noise form requires extra restrictions. A general f_i may be nonlinear in the noise and need not induce a Gaussian conditional distribution. Gaussian additive noise gives conditional variance equal to the noise variance; Gaussian linear SEMs give a jointly Gaussian vector.

4.1 The five-variable linear example

Assume $\varepsilon_1, \dots, \varepsilon_5$ are IID $N(0,1)$ and set

$$X_1 = \varepsilon_1, \tag{27}$$
$$X_2 = \varepsilon_2, \tag{28}$$
$$X_3 = 2X_1 + X_2 + \varepsilon_3, \tag{29}$$
$$X_4 = X_1 + 5X_2 + \varepsilon_4, \tag{30}$$
$$X_5 = X_2 + X_4 + \varepsilon_5. \tag{31}$$

This is a separate DAG from the earlier generative example. In matrix notation,

$$X = \Gamma X + \varepsilon, \qquad \Gamma = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 \\ 1 & 5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}. \tag{32}$$

Rows index child equations and columns index parents; thus Γ_{ij} is the coefficient on X_j in the equation for X_i. This is the transpose direction of the earlier adjacency convention $E_{ij} = 1$ for $i \to j$. Since Γ is strictly lower triangular, $I - \Gamma$ is invertible and

$$\varepsilon = (I - \Gamma)X \sim N_5(0, I_5), \tag{33}$$
$$X = (I - \Gamma)^{-1}\varepsilon = A\varepsilon, \tag{34}$$
$$\mathbb{E}[X] = 0, \qquad \Sigma = \mathrm{Cov}(X) = AA^T, \qquad A = (I - \Gamma)^{-1}. \tag{35}$$

Supplement: Explicit covariance and Cholesky factor

Substitution gives

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 \\ 1 & 5 & 0 & 1 & 0 \\ 1 & 6 & 0 & 1 & 1 \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 1 & 5 & 6 \\ 2 & 1 & 6 & 7 & 8 \\ 1 & 5 & 7 & 27 & 32 \\ 1 & 6 & 8 & 32 & 39 \end{pmatrix}. \tag{36}$$

The lower triangular matrix A has positive diagonal, so $\Sigma = AA^T$ is its Cholesky factorization, as suggested by the handwritten note. More generally, independent noises with covariance D give

$$\Sigma = (I - \Gamma)^{-1} D (I - \Gamma)^{-T}. \tag{37}$$

For Gaussian noise, the precision matrix is

$$\Sigma^{-1} = (I - \Gamma)^T D^{-1} (I - \Gamma). \tag{38}$$

This connects the SEM to Gaussian graphical models in the first note. Parent–child and co-parent pairs can contribute precision entries; coefficients may cancel, so a moral edge need not imply a nonzero precision entry for every parameter choice.